# Paper I – Basic Sciences in Urology (MCh Urology, Summer 2017)

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## I-1. Discuss anatomy of kidney. Describe role of nephron sparing surgery in small renal mass.

**Type:** LAQ (25 marks)

### Surgical anatomy of the kidney

- **Size and position:** about 11 cm, 125–170 g; T12–L3 on psoas, so the upper pole is more medial and posterior. The right kidney is 1–2 cm lower (liver).
- **Relations:**

| Aspect | Right | Left |
|---|---|---|
| Superior | Liver, adrenal, 12th rib | Spleen, stomach, adrenal, 11th–12th ribs |
| Medial/anterior | Duodenum (2nd part), hepatic flexure | Pancreatic tail, splenic vessels, splenic flexure, jejunum |
| Posterior (both) | Diaphragm, psoas, quadratus lumborum, subcostal nerve | Same |

- **Coverings (inside out):** true capsule, perirenal fat, **Gerota fascia**, pararenal fat. Gerota fascia is closed above and laterally but open below. So it contains tumour and abscess, but fluid can track into the pelvis.
- **Hilum (anterior to posterior):** renal vein, artery, pelvis, then the posterior segmental artery.
- **Arteries:** from the aorta at L1–L2; the right passes behind the IVC (inferior vena cava). Five **segmental end arteries** (apical, upper, middle, lower from the anterior division; posterior). With no anastomoses, occlusion causes segmental infarction, but each segment can be resected separately. Then interlobar (columns of Bertin), arcuate, interlobular arteries. Accessory arteries in 25–28%.
- **Brödel line:** an avascular plane between anterior and posterior territories, just behind the lateral border. Incision here bleeds least (anatrophic nephrolithotomy).
- **Veins** anastomose freely, so a segmental vein can be ligated safely. The long left renal vein receives adrenal, gonadal and lumbar veins.
- **Parenchyma:** cortex and 8–18 pyramids; 0.4–1.2 million nephrons per kidney.
- **Collecting system:** papilla, minor calyces (7–9), major calyces, pelvis. Puncture through the **calyceal fornix**, not the infundibulum (Sampaio).

- **Also know:** lymph to para-aortic (left) or paracaval/interaortocaval (right) nodes; sympathetic T8–L1.

**Diagram to draw:** segmental arteries with the Brödel line on a transverse kidney section.

## Small renal mass (SRM)

- **Definition:** an enhancing renal mass ≤4 cm (cT1a), usually incidental.
- **Biology:** 20–30% are benign (oncocytoma, fat-poor angiomyolipoma); most cancers are indolent (<2% metastasise).
- **Rationale:** radical nephrectomy (RN) causes CKD (chronic kidney disease), which raises cardiovascular events and death. PN (partial nephrectomy) gives equal cancer control with better function.

## Work-up

- Multiphase CT (computed tomography) or MRI (magnetic resonance imaging); chest imaging; eGFR (estimated glomerular filtration rate) and proteinuria. **Renal mass biopsy** if it may change management (non-diagnostic about 14%).
- **Nephrometry scores** predict complexity and complications:

| R.E.N.A.L. component | Scored as |
|---|---|
| **R**adius | ≤4, 4–7, ≥7 cm |
| **E**xophytic/endophytic | ≥50% exophytic, <50%, entirely endophytic |
| **N**earness to collecting system | ≥7, 4–7, ≤4 mm |
| **A**nterior/posterior | Descriptor |
| **L**ocation to polar lines | Polar, crossing, central |

- Total: low 4–6, moderate 7–9, high 10–12. **PADUA** and C-index are alternatives.

## Indications for nephron-sparing surgery

| Category | Situations |
|---|---|
| Imperative | Solitary kidney, bilateral tumours, RN would mean dialysis |
| Relative | Contralateral kidney threatened (stone, stenosis, diabetes), hereditary RCC (renal cell carcinoma) |
| Elective | cT1a with a normal contralateral kidney: **standard of care** |

## Options (AUA (American Urological Association) 2021)

- **Partial nephrectomy: preferred for cT1a.** Open, laparoscopic or robotic (RAPN, robot-assisted PN) with equal oncology. Steps:
  1. Expose kidney; intraoperative ultrasound marks the tumour.
  2. Hilar control; clamp artery (± vein), selective or off-clamp.
  3. Excise with a thin rim (or enucleate); negative margin is the priority.
  4. Close collecting system and vessels; renorrhaphy (sliding clip); unclamp; drain.
- **Ischaemia:** warm ischaemia traditionally <25–30 min; ice slush if longer is expected. Function depends mainly on **parenchyma preserved**; randomised trials show no lasting benefit of off-clamp.
- **Trifecta:** negative margin, no major complication, warm ischaemia ≤25 min.
- **Thermal ablation** (cryoablation, radiofrequency): masses <3 cm, percutaneous, biopsy first; higher local recurrence, salvaged by repeat ablation.
- **Active surveillance:** <2 cm, elderly or comorbid. Intervene if >3 cm or growth >5 mm a year (DISSRM registry: almost no metastasis).

- **RN** only if all three: high complexity, no CKD or proteinuria, and post-RN eGFR likely >45.

## Complications of PN

- Urine leak (stent), bleeding and pseudoaneurysm (angioembolisation), arteriovenous fistula, acute kidney injury, positive margin (0–7%; surveillance), hyperfiltration proteinuria.

## Evidence

- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** RN vs PN for ≤5 cm tumours: PN better function, similar oncology; overall survival slightly better with RN, but the trial is flawed.
- **Also know:** von Hippel–Lindau: treat at 3 cm, enucleation; 3D models.

> **Key points to remember:**
>
> - Five segmental end arteries; Brödel line avascular; puncture at the fornix.
> - SRM ≤4 cm: 20–30% benign; PN is standard, ablation <3 cm, surveillance <2 cm.
> - R.E.N.A.L. 4–6 / 7–9 / 10–12; trifecta includes warm ischaemia ≤25 min.
> - EORTC 30904: PN better function, similar cancer control.

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# I-2. Describe Pathophysiology of urosepsis and its management.

**Type:** LAQ (25 marks)

## Definitions

- **Urosepsis:** sepsis whose source is the urogenital tract. It is usually Gram-negative and usually follows obstruction or instrumentation.
- **Sepsis-3 (2016):**

| Term | Definition |
|---|---|
| **Sepsis** | Life-threatening organ dysfunction from a dysregulated host response to infection: **SOFA (Sequential Organ Failure Assessment) rise ≥2** |
| **qSOFA (quick SOFA)** | Respiratory rate ≥22, altered mentation, systolic blood pressure ≤100 mmHg; ≥2 = high risk |
| **Septic shock** | Vasopressor for MAP (mean arterial pressure) ≥65 mmHg **and** lactate >2 mmol/L despite fluids |

- SIRS (systemic inflammatory response syndrome) criteria were dropped as non-specific.

## Causes and risk factors

- **Obstruction with infection:** stone, PUJ (pelviureteric junction) obstruction, tumour, pregnancy. Pyonephrosis, emphysematous pyelonephritis, renal, perinephric and prostatic abscess.
- **Instrumentation:** PCNL (percutaneous nephrolithotomy), ureteroscopy with high irrigation pressure, TURP (transurethral resection of prostate), prostate biopsy, catheters.
- **Host:** diabetes, immunosuppression, old age, neurogenic bladder.
- **Organisms:** *E. coli* (commonest), *Proteus*, *Klebsiella*, *Pseudomonas*, *Enterococcus*; ESBL (extended-spectrum β-lactamase) producers rising; *Candida* in catheterised diabetics.

## Pathophysiology

### 1. Entry of bacteria into blood

- Obstruction causes stasis, impairs washout and raises pelvic pressure. Bacteria and endotoxin then enter the blood by **pyelovenous and pyelolymphatic backflow**.
- High irrigation pressure at endoscopy does the same, and so does fragmenting infected stones, which release endotoxin.
- An obstructed infected kidney is an **abscess with a collecting system**. Antibiotics barely reach it, and bladder urine may be sterile.

### 2. Host response

```
Endotoxin (LPS, lipopolysaccharide) / Gram-positive wall products
  → TLR4 (Toll-like receptor 4) on macrophages → NF-κB (nuclear factor κB)
  → cytokines TNF-α (tumour necrosis factor), interleukins IL-1, IL-6, IL-8
  → complement + coagulation activation + endothelial injury
```

- **Vasodilatation:** inducible nitric oxide causes low systemic vascular resistance (warm shock).
- **Capillary leak:** glycocalyx damage leads to hypovolaemia and tissue oedema.
- **Coagulopathy:** tissue factor release causes microthrombi, platelet consumption and DIC (disseminated intravascular coagulation).
- **Myocardial depression** and **mitochondrial dysfunction:** cells cannot use oxygen, so lactate rises.

- **Immunoparalysis:** a later anti-inflammatory phase predisposes to secondary infection.

**3. Organ dysfunction:** AKI (acute kidney injury), ARDS (acute respiratory distress syndrome), encephalopathy, liver dysfunction, DIC, multi-organ failure.

## Clinical features

- Fever or hypothermia, rigors, flank pain, tachycardia, tachypnoea, confusion, oliguria, hypotension. Often starts within hours of an endourological procedure.

## Investigations

- **Lactate**, blood cultures ×2, urine culture (and pelvic urine at drainage), blood count, creatinine, bilirubin, platelets, coagulation, arterial gas, glucose. Procalcitonin helps decide when to stop antibiotics.
- **Imaging:** bedside ultrasound (hydronephrosis, debris); **CT (computed tomography)** for stone, gas, abscess.

## Management

### A. Resuscitation and antibiotics (Surviving Sepsis Campaign 2021)

```
Lactate + cultures → IV (intravenous) antibiotics within 1 hour
→ 30 mL/kg crystalloid if hypotensive or lactate ≥4 mmol/L
→ noradrenaline for MAP ≥65; add vasopressin
→ refractory shock: hydrocortisone 200 mg/day
→ ICU (intensive care unit); re-measure lactate
```

- **Empirical antibiotics** by local resistance: third-generation cephalosporin or piperacillin–tazobactam ± aminoglycoside; carbapenem if ESBL risk. Adjust for renal function.

### B. Source control: the key urological step

- **Drain the obstructed infected kidney urgently:** retrograde **double-J stent** or **percutaneous nephrostomy (PCN)**. Randomised trials show equal effect on sepsis.
  - Stent: suits coagulopathy; avoid high-pressure contrast injection.
  - PCN: better for large or impacted stones, pyonephrosis; allows pelvic urine culture.
- **No definitive stone surgery during sepsis.** Treat the stone after the infection clears.
- Other sources: drain renal or perinephric abscess; emphysematous pyelonephritis (drainage, nephrectomy if diffuse gas); prostatic abscess drainage; change or remove catheter; debride Fournier gangrene.

### C. Supportive care

- Oxygen and ventilation, glucose control, transfusion below haemoglobin 7 g/dL, thromboprophylaxis, stress-ulcer prophylaxis, renal replacement if needed.

### D. De-escalation

- Narrow antibiotics by culture at 48–72 h. Usual course 7–10 days; longer with an undrained focus. Step down to oral therapy when stable.

## Special situations

- **Emphysematous pyelonephritis** (diabetics): fluids, glucose control, antibiotics, drain; nephrectomy for diffuse gas without fluid.
- **Pregnancy:** stent or nephrostomy; avoid fluoroquinolones.
- **Fungal urosepsis:** remove catheter, antifungal, drain fungal balls.
- **Prognosis:** urosepsis does better than sepsis from other sites if the source is drained early; delay in drainage is the main avoidable cause of death.

## Prevention

- Sterile urine before elective endourology; antibiotic prophylaxis; low irrigation pressure and access sheath; short operating time with infected stones; stage the procedure if pus is found.

**Diagram to draw:** obstructed kidney leading to backflow, cytokine cascade and shock.

> **Key points to remember:**
>
> - Sepsis = SOFA ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2.
> - Obstruction plus infection is the emergency; bladder urine may be sterile.
> - Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline.
> - Stent and nephrostomy equally effective; delay definitive stone surgery.

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# I-3a. Complications of intravenous contrast media.

**Type:** SAQ (10 marks)

## Introduction

- Agents are tri-iodinated benzene rings, filtered by the glomerulus. Modern practice uses **non-ionic low-osmolar** (iohexol, about 600–850 mOsm/kg) or **iso-osmolar** (iodixanol, 290) agents. High-osmolar ionic agents caused far more reactions.

## Complications

| Type | Features |
|---|---|
| **Immediate hypersensitivity** (within 1 hour) | Urticaria, angio-oedema, bronchospasm, hypotension, anaphylaxis. Mostly non-IgE (immunoglobulin E) mast-cell release. Severe: 0.22% ionic vs 0.04% non-ionic |
| **Non-immediate** (hours to days) | Maculopapular rash; T-cell mediated |
| **Physiological** | Warmth, metallic taste, nausea, vasovagal reaction, arrhythmia, seizures |
| **CA-AKI (contrast-associated acute kidney injury)** | Creatinine rise ≥0.3 mg/dL in 48 h or ≥1.5 × baseline. Afferent vasoconstriction causes medullary hypoxia, plus direct tubular toxicity. Real risk mainly at eGFR (estimated glomerular filtration rate) <30 |
| **Metformin** | Lactic acidosis only if AKI develops |
| **Thyroid** | Hyperthyroidism in Graves' or nodular goitre; blocks radio-iodine for weeks |
| **Extravasation** | Usually minor; rarely compartment syndrome |

- **Risk factors:** previous reaction (strongest), asthma, atopy, cardiac disease; for CA-AKI: CKD (chronic kidney disease), diabetes, volume, intra-arterial route. **Shellfish allergy is not a risk.**

## Prevention

- Use the lowest dose of low- or iso-osmolar agent; consider ultrasound, MRI (magnetic resonance imaging) or non-contrast CT (computed tomography).
- **Premedication** (prior reaction): prednisone 50 mg at 13, 7 and 1 h, or methylprednisolone 32 mg at 12 and 2 h, plus diphenhydramine 50 mg 1 h before. Breakthrough reactions still occur.
- **CA-AKI:** IV (intravenous) isotonic saline for eGFR <30. **PRESERVE (2018):** bicarbonate and NAC (N-acetylcysteine) no benefit.
- **Metformin:** withhold 48 h only if eGFR <30 or AKI.

## Treatment of reactions

```
Stop injection → airway, oxygen, help
Urticaria → observe ± antihistamine
Bronchospasm → salbutamol inhaler
Anaphylaxis → IM (intramuscular) adrenaline 0.3–0.5 mg (1 mg/mL), repeat 5–15 min; IV fluids
Vasovagal → legs up, fluids, atropine
```

**Diagram to draw:** monomer versus dimer on an osmolality scale.

**Key points to remember:**

- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg.
- CA-AKI: saline if eGFR <30; bicarbonate and NAC useless.

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# I-3b. Flexible Ureterorenoscopy

**Type:** SAQ (10 marks)

## Introduction

- Flexible ureterorenoscopy (fURS) or RIRS (retrograde intrarenal surgery) inspects and treats the whole upper tract retrogradely. Marshall reported it in 1964; active deflection and a working channel made it practical.

## Instrument

| Feature | Details |
|---|---|
| Tip | 4.5–7.5 Fr, tapered (digital tips up to about 8.5 Fr) |
| Working channel | 3.6 Fr (laser fibre, basket) |
| Deflection | ≥270° both ways; lower pole needs about 140° (up to 175°) |
| Types | **Fibreoptic** (smallest, image degrades); **digital** (chip on tip, better image, not more durable); **single-use** (LithoVue 2016, 7.7 Fr; cheaper below about 99 cases a year) |

- Repair is needed about every 15 cases.

## Indications

- **Stones:** renal <2 cm; lower pole <1 cm (or >1 cm as an alternative to PCNL, percutaneous nephrolithotomy); patients who cannot stop anticoagulants; failed SWL (shock wave lithotripsy); obesity.
- **Others:** diagnosis and ablation of low-grade UTUC (upper tract urothelial carcinoma); filling defects, positive cytology; endopyelotomy, short strictures; calyceal diverticulum; essential haematuria; foreign bodies.

## Technique

1. Sterile urine; single-dose antibiotic.
2. Cystoscopy; **safety guidewire**; retrograde pyelogram.
3. **Access sheath** (usually 12/14 Fr) over a stiff wire under fluoroscopy: lowers intrarenal pressure, allows repeated passes; severe ureteric injury in about 1 in 8.
4. Low-pressure irrigation; **Ho:YAG (holmium:yttrium-aluminium-garnet) or thulium fibre laser**: dusting or fragment-and-basket. Keep power ≤10 W in ureter, ≤25 W in kidney (tissue damage above 43 °C).
5. Inspect ureter on withdrawal; stent unless uncomplicated.

- **Difficult lower pole:** steep infundibulopelvic angle, long (>10 mm) or narrow (<5 mm) infundibulum (EAU (European Association of Urology) factors against SWL; they also hinder fURS). Relocate the stone to an upper calyx with a basket.

## Complications

- Sepsis about 5%; stricture about 3%; bleeding or subcapsular haematoma <1%; perforation; avulsion <1%; stent symptoms; scope damage.

## Recent advances

- Suction access sheaths, intrarenal pressure sensors, thulium laser, robotic fURS.

**Diagram to draw:** flexible scope deflected into the lower pole calyx through an access sheath.

> **Key points to remember:**
> - 3.6 Fr channel; ≥270° deflection.
> - Access sheath lowers pressure but injures ureter in about 1 in 8.
> - Laser power ≤10 W ureter, ≤25 W kidney.

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## I-3c. Perinephric abscess

**Type:** SAQ (10 marks)

### Definition

- Pus **outside the renal capsule but within Gerota fascia**. Rupture through Gerota fascia gives a **paranephric** abscess.

### Aetiology

- Rupture of a renal cortical abscess or pyonephrosis (commonest); extravasation of infected urine from obstruction (stone).
- **Haematogenous** spread in about a third (skin source, *S. aureus*).
- Infected perirenal haematoma or urinoma (trauma, surgery).
- Spread from bowel (Crohn disease, perforation), pancreas, spine.
- Risk factors: **diabetes (about a third)**, stones, obstruction, polycystic kidneys on dialysis.
- Organisms: *E. coli*, *Proteus*, *S. aureus*; anaerobes with bowel sources.

### Clinical features

- **Insidious:** symptoms >5 days in most; more than a third afebrile.
- Flank pain, tenderness, flank mass (about half); psoas sign (flexed, externally rotated hip).
- **Suspect it when "pyelonephritis" persists:** pyelonephritis defervesces within 4–5 days; perinephric abscess does not.

| | Pyelonephritis | Perinephric abscess |
|---|---|---|
| Symptoms before admission | ≤5 days | >5 days |
| Fever on antibiotics | Never >4 days | ≥5 days (median 7) |

### Investigations

- Leucocytosis, raised creatinine, pyuria; urine culture identifies the organism in only about a third; blood cultures.
- **CT (computed tomography): investigation of choice.** It shows the collection, its spread (psoas, flank), the cause (stone, obstruction) and plans drainage.
- Ultrasound: quick; guides drainage. Plain X-ray: loss of psoas shadow, scoliosis.

### Management

- **Antibiotics immediately:** aminoglycoside plus antistaphylococcal penicillin; cephalosporin or vancomycin if allergic. Adjust to culture.
- **Drainage:** small (<3 cm) in an immunocompetent patient may resolve with antibiotics. Larger or non-responding: **CT- or ultrasound-guided percutaneous drainage** (first line); open drainage for loculated or failed cases.
- **Treat the cause:** relieve obstruction (stent or nephrostomy); **nephrectomy** for a non-functioning, pyonephrotic kidney, at the same time or staged.
- Mortality fell from 40–50% to about 12% with CT; delay in diagnosis causes most deaths.

**Diagram to draw:** coronal kidney showing renal, perinephric and paranephric spaces.

> **Key points to remember:**
> - Within Gerota fascia; diabetes in a third.
> - Fever beyond 4–5 days of antibiotics: get a CT.
> - Percutaneous drainage first line; nephrectomy if non-functioning.

## I-3d. Ciprofloxacin

**Type:** SAQ (10 marks)

### Class and mechanism

- Second-generation **fluoroquinolone**, derived from nalidixic acid.
- Inhibits **DNA gyrase** (topoisomerase II, main target in Gram-negatives) and **topoisomerase IV** (Gram-positives). This blocks DNA supercoiling and replication: **bactericidal**.

### Spectrum

- Enterobacteriaceae, ***Pseudomonas aeruginosa***, *Haemophilus*, *S. saprophyticus*, *S. aureus* (methicillin-sensitive); *Legionella*, some mycobacteria.
- Weak against streptococci and enterococci; **no anaerobic cover**, so bowel and vaginal flora are partly spared.

### Pharmacokinetics

- Oral bioavailability about 70%; IV (intravenous) and oral forms; half-life about 4 h.
- **Concentration-dependent killing:** AUC/MIC (area under curve to minimum inhibitory concentration) predicts effect; post-antibiotic effect.
- High urine and tissue levels; **good prostatic penetration**.
- Renal excretion: reduce dose if creatinine clearance <30 mL/min.

### Doses

- Cystitis 250 mg twice daily × 3 days; pyelonephritis 500 mg twice daily × 7 days (or 1 g extended-release daily); IV 400 mg 12-hourly.
- Chronic bacterial prostatitis: 4–6 weeks.

### Indications and restrictions

- Pyelonephritis and complicated UTI (urinary tract infection) **only where local resistance ≤10%**; prostatitis; prophylaxis for transrectal biopsy (falling, because of resistance).
- **Not first line for uncomplicated cystitis** (FDA (US Food and Drug Administration) 2016; European restrictions 2019): use only when no alternative.

### Resistance

- *gyrA*/*parC* mutations, efflux pumps, reduced porins, plasmid *qnr* genes. Linked to ESBL (extended-spectrum β-lactamase) *E. coli* (ST131). Main cause of post-biopsy sepsis.

### Adverse effects

- GI (gastrointestinal) upset (commonest), photosensitivity, *C. difficile*.
- **FDA boxed warnings:** tendinopathy and **Achilles rupture** (2008; risk higher with age, steroids, renal failure); peripheral neuropathy (2013); disabling effects (2016); psychiatric effects and dysglycaemia (2018); aortic aneurysm and dissection (2018).
- CNS (central nervous system): seizures, confusion; QT prolongation.

### Interactions

- Antacids, iron, zinc, calcium block absorption (separate by ≥2 h). Raises theophylline and tizanidine levels; potentiates warfarin.

### Pregnancy and children

- Avoid: **cartilage damage** in immature animals. Also avoid in breastfeeding.

**Key points to remember:**

- DNA gyrase and topoisomerase IV; concentration-dependent.
- Use for pyelonephritis only if local resistance ≤10%.
- Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

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# I-3e. Hassan's canula.

**Type:** SAQ (10 marks)

*Assumed to mean the Hasson cannula, used for open laparoscopic access.*

## Introduction

- Described by **Harrith Hasson (1971)** for open ("direct vision") laparoscopy. The first port is placed under vision instead of a blind Veress needle and trocar.
- **Parts:** outer sleeve with a cone-shaped (olive) adapter, **blunt obturator**, struts to anchor fascial stay sutures, and a gas inlet with tap.

## Indications

- **Previous abdominal surgery or expected adhesions** (main indication).
- Organomegaly, pregnancy, aortic aneurysm, children, thin patients.
- Retroperitoneoscopy and extraperitoneal access (open start, then balloon dilatation).

## Technique (transperitoneal)

1. 2-cm semicircular infra-umbilical incision.
2. Open fascia and then peritoneum separately, under vision, finger-sized.
3. Confirm entry by sight and finger sweep (free adhesions).
4. Place two 0 silk fascial stay sutures.
5. Insert the cannula with its **blunt obturator**.
6. Seat the cone in the defect; wrap stay sutures round the struts to seal.
7. Remove obturator; insufflate at high flow (to 12–15 mmHg).
8. At closure, tie the stay sutures to close the fascia.

- **Retroperitoneal:** 2–2.5 cm incision below the 12th rib tip, finger dissection onto psoas, balloon dilatation, then port.

## Advantages over Veress

- No blind puncture: fewer major vessel and bowel injuries; no preperitoneal insufflation; no gas embolism from intravascular insufflation.
- Fast insufflation; easy fascial closure (less port-site hernia).

## Disadvantages

- Larger incision; **gas leak** around the port; slightly longer in obese patients.
- Early retroperitoneal use caused subcutaneous emphysema and hypercarbia, now solved by balloon-cuffed blunt ports.

## Complications

- Bowel injury if adherent under the wound (repair through the same incision); wound infection; port-site hernia; vascular injury is still possible.

## Evidence

- Reviews show no single entry method is clearly safest for major injury; open entry has fewer failed entries.

**Diagram to draw:** Hasson cannula with cone, stay sutures and blunt obturator in the abdominal wall.

**Key points to remember:**

- Hasson 1971: open entry under vision.
- Main indication: previous surgery or adhesions.
- Avoids blind Veress injuries; drawback is gas leak.

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# I-3f. Proteinuria

**Type:** SAQ (10 marks)

## Definition

- Normal protein excretion is **<150 mg/day** (albumin <30 mg/day). The filter blocks albumin and the proximal tubule reabsorbs filtered small proteins.
- Normal mix: about 30% albumin, 30% globulins, 40% tissue proteins (Tamm–Horsfall).
- **Nephrotic range:** >3.5 g/day, with hypoalbuminaemia, oedema and hyperlipidaemia.

## Quantification (KDIGO, Kidney Disease: Improving Global Outcomes)

| Category | ACR (albumin-to-creatinine ratio) | PCR (protein-to-creatinine ratio) | 24-h protein |
|---|---|---|---|
| A1 normal–mild | <30 mg/g (<3 mg/mmol) | <150 mg/g (<15 mg/mmol) | <150 mg |
| A2 moderate | 30–300 mg/g | 150–500 mg/g | 150–500 mg |
| A3 severe | >300 mg/g (>30 mg/mmol) | >500 mg/g | >500 mg |

- **Spot early-morning ACR is first line**; PCR if non-albumin protein suspected; 24-h urine with creatinine check.
- **Dipstick** (tetrabromophenol blue) detects albumin only, from 20–30 mg/dL; misses light chains. **Sulfosalicylic acid** detects all proteins.

## Types

| Type | Mechanism | Amount | Example |
|---|---|---|---|
| **Glomerular** (commonest) | Leaky filter | Often >2 g; albumin ≈70% | Diabetes, glomerulonephritis, FSGS (focal segmental glomerulosclerosis) |
| **Tubular** | Failed reabsorption of small proteins | <2 g; mainly globulins | Fanconi syndrome, heavy metals, drugs |
| **Overflow** | Excess plasma protein | Variable | Myeloma (Bence Jones), myoglobin |

- **Transient:** fever, exercise, heart failure. **Orthostatic:** young men, upright only, <1 g; benign. Test with a split (supine overnight vs upright day) collection.

## Evaluation

```
Positive dipstick → repeat; exclude UTI (urinary tract infection), fever, exercise
→ persistent → ACR/PCR or 24-h protein
→ creatinine/eGFR (estimated glomerular filtration rate), sediment (dysmorphic red cells,
casts), ultrasound
→ glomerular pattern → nephrology ± renal biopsy
→ globulins → immunofixation (myeloma)
```

## Urological relevance

- **Haematuria work-up:** proteinuria, casts or dysmorphic cells suggest medical renal disease; urological bleeding rarely raises protein much.
- **Reflux nephropathy:** proteinuria marks progression to renal failure.
- **After nephrectomy or partial nephrectomy:** >50% loss of mass causes hyperfiltration, proteinuria and FSGS. AUA (American Urological Association) 2021 counts proteinuria as a reason to prefer nephron sparing and refer to nephrology.
- **Living donors:** significant proteinuria excludes donation.

## Management

- Treat the cause; blood pressure control; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)**; SGLT2 (sodium–glucose cotransporter-2) inhibitors.

**Key points to remember:**

- Normal <150 mg/day; nephrotic >3.5 g/day.
- ACR A2 30–300, A3 >300 mg/g.
- Dipstick misses light chains; orthostatic proteinuria is benign.

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# Rapid Revision Sheet – Paper I

*All the key points from this paper in one place. Read this in the last 30 minutes before the exam.*

**I-1. Discuss anatomy of kidney. Describe role of nephron sparing surgery in small renal mass. (LAQ)**

- Five segmental end arteries; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard, ablation <3 cm, surveillance <2 cm.
- R.E.N.A.L. 4–6 / 7–9 / 10–12; trifecta includes warm ischaemia ≤25 min.
- EORTC 30904: PN better function, similar cancer control.

**I-2. Describe Pathophysiology of urosepsis and its management. (LAQ)**

- Sepsis = SOFA ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2.
- Obstruction plus infection is the emergency; bladder urine may be sterile.
- Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline.
- Stent and nephrostomy equally effective; delay definitive stone surgery.

**I-3a. Complications of intravenous contrast media. (SAQ)**

- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg.
- CA-AKI: saline if eGFR <30; bicarbonate and NAC useless.

**I-3b. Flexible Ureterorenoscopy (SAQ)**

- 3.6 Fr channel; ≥270° deflection.
- Access sheath lowers pressure but injures ureter in about 1 in 8.
- Laser power ≤10 W ureter, ≤25 W kidney.

**I-3c. Perinephric abscess (SAQ)**

- Within Gerota fascia; diabetes in a third.
- Fever beyond 4–5 days of antibiotics: get a CT.
- Percutaneous drainage first line; nephrectomy if non-functioning.

**I-3d. Ciprofloxacin (SAQ)**

- DNA gyrase and topoisomerase IV; concentration-dependent.
- Use for pyelonephritis only if local resistance ≤10%.
- Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

**I-3e. Hassan's canula. (SAQ)**

- Hasson 1971: open entry under vision.
- Main indication: previous surgery or adhesions.
- Avoids blind Veress injuries; drawback is gas leak.

**I-3f. Proteinuria (SAQ)**

- Normal <150 mg/day; nephrotic >3.5 g/day.
- ACR A2 30–300, A3 >300 mg/g.
- Dipstick misses light chains; orthostatic proteinuria is benign.